

Machine Learning for Quants

From Signals to Decisions, Not Accuracy Scores

Amit Kumar Jha, Quant Analyst at UBS

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The 3-Second Answer (Module 12)

Question: "Your model predicts Up with 51% probability. Do you trade?"

Your opening line:

"Likely no. The edge is only 1% (2% if round trip), which is likely below transaction costs. I only trade when the confidence exceeds a threshold, say $p > 0.55$, to ensure the expected PnL covers execution slippage and fees."

Why this wins:

- Calculates the economic reality (1% edge < costs).
- Introduces the concept of thresholds (Cost-aware).
- Shows profitability focus (Not just accuracy).

Follow-up ready:

- "Threshold level?" -> Set by backtesting PnL including costs.
- "If 0.6?" -> Yes, size aggressively.

Red flag to avoid:

Never say: "Yes, 51% is better than random" (It's a losing strategy after costs).

Kelly Sizing Variants

1. Full Kelly (Aggressive):

$$f^* = \frac{p \cdot b - q}{b} = \frac{p \cdot b - (1 - p)}{b}$$

2. Half Kelly (Conservative - Recommended):

$$f^* = \frac{1}{2} \times \frac{p \cdot b - q}{b}$$

3. ML-Calibrated Kelly:

$$f^* = \frac{(p - 0.5) \times \text{Sharpe}}{2} \times \text{Calibration Factor}$$

Warning Signs:

- If $f^* > 0.25 \rightarrow$ Model overconfident, cap at 25%
- If $p < 0.55 \rightarrow$ Edge too small after costs, size = 0

Module 13: Crisis Mode: ML Edition

Module Overview

What this module is about: What to do when the ML model explodes.

Why this matters: ML models react violently to data points outside their training manifold (Out of Distribution - OOD).

What confusion this clears:

- Out of Distribution (OOD) detection.
- Kill-switch logic.
- Fallback rules.

After this module, you should be able to answer: "How do you detect when your model is in a crisis?"

13.1 Crisis Triggers

1. **Accuracy Collapse:** Rolling accuracy drops below 50% (Random) for 5 days.
2. **Feature Drift:** KS Test on inputs $p < 0.01$ (Massive shift).
3. **Correlation Spike:** Diversification vanishes (Market meltdown).

32. "How do you handle imbalanced data in credit risk?"

Question: Class weights vs threshold tuning.

Answer: "I never upsample/downsample—that leaks info. Instead: 1) Use class weights inversely proportional to default frequency. 2) Tune decision threshold on validation AUC-PR, not accuracy. 3) Report Expected Loss = PD * LGD * EAD, not just PD. 4) Out-of-time validation is mandatory—can't shuffle months in credit cycles."

33. "Walk me through your model validation pipeline."

Question: End-to-end rigor.

Answer: "Three nested loops: Inner loop tunes hyperparameters via Bayesian optimization on rolling 6-month train sets. Middle loop evaluates on subsequent 2-month validation sets, tracking Sharpe decay. Outer loop runs walk-forward for 5 years. Test set is 2022-2023 crisis data, touched once. If OOS Sharpe < 1.0, model is killed. If drift detected, retrain window shrinks from 2 years to 6 months."

34. "Why does ensemble diversification fail in crises?"

Question: Correlation → 1 problem.

Answer: "During normal times, my equity value model and momentum model have $\rho=0.3$. In March 2020, both loaded on 'liquidation'—value because cheap stocks got cheaper, momentum because winners reversed. The learned features weren't orthogonal; they were both betting on market stability. True diversification requires different data sources, not just different algorithms on same data."

35. "How would you detect if a feature is leaked?"

Question: Precision debugging.

Answer: "I'd run a ghost model using ONLY that feature. If its standalone accuracy is >60%, it's leaking future info. Second test: remove the feature and see if ensemble accuracy drops less than its feature importance suggests—if so, it was carrying information that other features also had (redundancy). Third: check timestamp alignment with a fine-tooth comb—especially around corporate actions."

One-Pager Cheat Sheets (Print and Laminate)

Card 1: Crisis Mode Activation (ML Edition)

CRISIS MODE - ACTIVATE IMMEDIATELY

Triggers (Any 1 = Crisis):

- ☐ Feature Distribution KS-Test $p < 0.01$
- ☐ Model Accuracy < 52% (Random)
- ☐ Prediction Confidence > 95% (Hallucination)
- ☐ Correlation Matrix Explodes ($\rho \rightarrow 1$)
- ☐ OOD (Out of Distribution) Detected

Actions (Execute within 5 mins):

1. **DISABLE:** Turn off ML model inputs immediately.
2. **FALLBACK:** Switch to Risk-Off / Econometric Rules.
3. **FLATTEN:** Reduce gross exposure to minimum.
4. **AUDIT:** Run diagnostics on feature pipeline.

Do NOT:

- Trust high probabilities in a crash (Likely hallucinated).
- Re-tune hyperparameters live (Will overfit to crash).

Mental Model 4: “If you tuned it, you touched it”

Validation set performance is optimistic if you iterated on it. Real performance is always lower than the best validation score.

Final Interview Checklist — ML for Quants

60-Second Self-Test

- ☐ Can explain the Bias-Variance tradeoff?
- ☐ Can implement Walk-Forward validation?
- ☐ Can distinguish Regression vs Classification targets?
- ☐ Can detect Overfitting (Drift, Leakage)?
- ☐ Can explain a Black Box (SHAP, Feature Imp)?
- ☐ Know when to use Trees vs Linear?
- ☐ Can handle Imbalanced data?
- ☐ Understand the risk of “Future Leakage”?

If you can't answer any of these in 60 seconds, reread that module.

ML vs Stats: The Final Verdict

- **Use Stats:** For inference, hypothesis testing, risk modeling, and when T is small. (Hypothesis → Decision).
- **Use ML:** For prediction, pattern recognition in high-dim data, and regime classification. (Data → Prediction).
- **Use Hybrid:** Stats for structure/hedges, ML for residuals/signal.

Interview: “I use Statistics to define the risk envelope, and Machine Learning to optimize PnL within that envelope.”